

DEEP SANDIP MEHTA

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Summary

Third-year Computer Science undergraduate (SPIT Mumbai) with active research at ISRO, Project Developer at SPJIMR and two national hackathon wins. Specializes in deep learning, production-grade retrieval-augmented generation, and full-stack machine learning system design.

Education

Bachelor of Technology, Computer Science and Engineering – Sardar Patel Institute of Technology, Mumbai *GPA: 8.75/10 | June 2028*

Minor Degree in Fintech – L&T Edutech *GPA: 10/10 | June 2028*

Technical Skills

Languages	Python, C++, JavaScript, SQL
Machine Learning	PyTorch, Scikit-learn, ONNX Runtime, Transformers, CNN, LSTM, RAG, Pandas, NumPy
Frameworks	FastAPI, React.js, Next.js, Node.js, Spring Boot, Tailwind CSS
Tools	PostgreSQL, MySQL, MongoDB, Docker, Git, Vercel

Experience

ML Research Intern | Indian Institute of Remote Sensing (IIRS), ISRO *June 2026 – Present*

- Developed a **multi-stage atmospheric data analysis pipeline** integrating 10 heterogeneous HDF4/HDF5/NetCDF datasets; implemented temporal alignment, climatology/trend removal, ENSO/IOD confounder removal, and AR(1) prewhitening. Applied **PCMCI+**, **Granger causality**, **Transfer Entropy**, **EOF**, **correlation**, and **bootstrap analyses** with Benjamini–Hochberg FDR correction to investigate aerosol–isotope and moisture–isotope relationships across multiple Indian regions.

Project Intern | S.P. Jain Institute of Management and Research (SPJIMR), Mumbai *June 2026 – Present*

- Built a **stock market simulation platform** (Spring Boot, React 18, MySQL, Docker) with a stochastic five-factor pricing engine (innovation, liquidity, sentiment) serving 500 NSE/BSE companies via STOMP WebSocket at sub-16ms delta latency.
- Developed dual-role dashboards (professor/student) with live candlestick charts, JWT + Spring Security RBAC, and a **retail investor education module** covering key financial ratios aligned to SEBI/AMFI classifications.

Projects

Pixie – Insurance Education Retrieval-Augmented Generation Chatbot *Mar – Apr 2026 [Code]*
Full-Stack Machine Learning Engineer | Production system serving Indian insurance consumers

- Architected a **hybrid RAG pipeline** combining BM25 + dense retrieval (BAAI/bge-large), Reciprocal Rank Fusion, cross-encoder reranking, and PageRank citation expansion over IRDAI regulatory documents, improving answer precision by **40%**.
- Built a self-evaluation loop with entity grounding and TF-IDF contextual compression, reducing irrelevant context by 40% and cutting hallucination rate; delivered multilingual synthesis across **13 Indian languages** via Groq LLaMA 3.3-70B at under 200ms to first token.

Aarogya Sampark – Rural Healthcare Decision Support System *Apr 2026 [Code]*
Full-Stack Machine Learning Engineer | AIRAVAT 3.0 Hackathon, IEEE Computer Society SPIT

- Built an **offline-first PWA** with a two-stage inference pipeline — ONNX Runtime Web for on-device inference and Llama-2-70B for online synthesis — enabling zero-connectivity clinical support for rural health workers.
- Deployed three WHO-grounded clinical agents (maternal hemorrhage, TB adherence, pesticide triage) across **6 regional languages**, reducing average triage time by an estimated **8 minutes per case**.

KrushUnnati – Smart Agriculture Platform *Jan 2026 [Code]*
Full-Stack and Machine Learning Developer | Track Winner, Quasar 4.0 National Hackathon

- Built Scikit-learn pipelines achieving **94% crop yield classification accuracy**; delivered real-time soil analytics to 150+ farmers, cutting manual monitoring effort by 35%.
- Architected a decentralized direct-to-consumer marketplace (React.js, Node.js), eliminating two intermediary layers and reducing supply chain overhead by an estimated **25%**.

Achievements

- Domain Winner, Quasar 4.0 National Hackathon** – 1st place, Smart Agriculture track, 100+ teams, VPPCOE (Jan 2026).
- 1st Place, IIT Hyderabad Blockchain Hackathon** – Decentralized app for financial transparency and secure fund disbursement (Jan 2026).
- Five-Time National Case Study Finalist** – Final rounds at premier engineering and business institutes across India.

Certifications

- Stanford / DeepLearning.AI** – Supervised ML, Advanced Learning Algorithms, Unsupervised Learning
- WorldQuant University** – Data Science Lab, Deep Learning Lab
- Scrimba** – Frontend Career Path | **GeeksforGeeks** – Product Management, Next.js | **SPIT** – C Programming